THE DIMENSION OF A FIBRE AT A POINT IS AT MOST THE SUM OF THE FIBRE DIMENSIONS ALONG THE COMPOSITE, WITH EQUALITY UNDER FLATNESS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 5395-5407 and proof span 5409-5422 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof refers to the fibre dimension formula of an earlier statement and to two algebra statements; the references are part of the counted span and are handled as navigation pointers.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The description of the dimension of a fibre at a point as the sum of the dimension of the local ring of the fibre and the transcendence degree of the residue field over the residue field of the base point, cited from the same chapter. Cited at proof lines 5411-5412. Reduces the geometric inequality to an inequality between dimensions of local rings of the two fibres.
- Additivity of the transcendence degree over a tower of field extensions, and the corresponding inequality and equality statements for dimensions of local rings of fibres over a base, cited from the algebra chapter. Used at proof lines 5409-5410 and 5414-5422. Supplies the field degree identity and the two local algebra statements that finish the proof.

Source reference. Original lemma 00ON: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 901–901. Same-chapter reference used in the retained author text at source line(s) 5421, 5699. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00OM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 900–900. Same-chapter reference used in the retained author text at source line(s) 5419. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02FX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3208–3208. Same-chapter reference used in the retained author text at source line(s) 5412, 5659, 5661, 13444. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to Y$ and $g : Y \to S$ be morphisms of schemes. Let $x \in X$ and set $y = f(x)$, $s = g(y)$. Assume f and g locally of finite type. Then*

$$\dim_x(X_s) \leq \dim_x(X_y) + \dim_y(Y_s).$$

Moreover, equality holds if $\mathcal{O}_{X_s,x}$ is flat over $\mathcal{O}_{Y_s,y}$, which holds for example if $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,y}$.

Proof. Note that $\mathrm{trdeg}_{\kappa(s)}(\kappa(x)) = \mathrm{trdeg}_{\kappa(y)}(\kappa(x)) + \mathrm{trdeg}_{\kappa(s)}(\kappa(y))$. Thus by Lemma 02FX the statement is equivalent to

$$\dim(\mathcal{O}_{X_s,x}) \leq \dim(\mathcal{O}_{X_y,x}) + \dim(\mathcal{O}_{Y_s,y}).$$

For this see Algebra, Lemma 00OM. For the flat case see Algebra, Lemma 00ON. □